



Faculty of Engineering

Department of Biomedical Engineering

Non-Contact Type Clinical Thermometer Verification/Validation Test Plan

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CHANGE RECORDS

FIRST RELEASE

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1. PURPOSE

The purpose of this document is to outline the formal engineering procedures and protocols required to verify and validate the design, hardware compilation, and firmware functionality of the Non-Contact Type Clinical Thermometer. This plan serves as the definitive structural guide for executing system evaluations, ensuring that the completed prototype complies entirely with the safety, performance, and operational boundaries defined in the System Requirements Document

2. SCOPE

The scope of this document encompasses the complete execution of verification and validation (V&V) testing methodologies applied directly to the prototype Non-Contact Type Clinical Thermometer (Part Number: 01-MIY-002). This includes testing the battery-powered hardware setup, the 3D-printed enclosure mechanics, the OLED display metrics, and the baseline tracking firmware.

V&V activities covered under this scope will be executed in an open laboratory setting using controlled heating/cooling plates, reference thermometers, and PC-based serial data loggers. A comprehensive Verification/Validation Matrix is included within this document to establish full traceability between each system requirement and its designated testing methodology.

3. REFERENCES

3.1. 01-MIY-002: System Requirements for Clinical IR Thermometer

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4. METHODS FOR VERIFICATION/VALIDATION

4.1 INSPECTION

Inspection refers to the visual, tactile, and dimensional examination of the hardware without actively operating the system. This method will be utilized to confirm physical compliance metrics, including:

- Verifying that the physical mass of the thermometer falls within the mandated 50g to 400g limits.
- Inspecting the 3D-printed housing for a smooth, cleanable, and defect-free surface finish.
- Manually checking that the user interface does not exceed the maximum limit of two operating buttons.
- Physically measuring the OLED display dimension to ensure it exceeds 0.96 inches.

4.2 ANALYSIS

Analysis encompasses the process of confirming whether a specification requirement has been met through thorough technical evaluation, mathematical calculations, and data processing rather than direct physical measurement alone. This method is utilized to evaluate complex system parameters and data.

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4.3. DEMONSTRATION

Demonstration involves a qualitative validation procedure aimed at showcasing the critical functions, user interactions, and visual capabilities of the system. Unlike rigorous testing, this process relies on clear pass/fail operational checks to exhibit usability and real-world performance, including:

- Operating the minimal, user-friendly control interface to prove that an individual can easily click and interact with the power and measurement switches.
- Observing the device behavior when left completely idle to demonstrate that the firmware successfully triggers an automatic power-saving or low-power sleep state within the designated timeline.
- Visually verifying the high-contrast output on the OLED screen to ensure that character sizes are larger than 4mm for easy reading and that the temperature values distinctly display the degrees Celsius metric

4.4. TEST

Testing involves a comprehensive, quantitative, and systematic examination of the thermometer under tightly controlled laboratory inputs. This methodology relies on calibrated measurement instruments and repeatable procedures to apply specific inputs, observe system responses, and meticulously evaluate the resulting data against strict numerical acceptance criteria.

- Formally validating that the thermometer successfully records temperatures across the target output range.
- Quantifying that the measurement acquisition time remains securely between 0 and 1 seconds.
- Subjecting the device to extreme environments inside an environmental chamber to verify operational stability and storage survival boundaries.

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4.5. VERIFICATION / VALIDATION MATRIX

The Verification/Validation Matrix contains the paragraph numbers and titles of requirement specifications, along with verification methods and procedures. Detailed instructions are provided within associated test procedures like the Acceptance Test Procedure (ATP) and Verification/Validation Test Procedure (V2TP). The creation of the Verification/Validation Test).

Verification/Validation Test Procedure will be created according to this plan.

System Requirement Specification paragraph 01-MIY-002	Requirement Title	Method	Procedure
4.1	Physical Requirements		
4.1.1	Weight	Inspection	V2TP
4.1.2	Surface	Demonstration	V2TP
4.2	Functional Requirements		
4.2.1	Operating Controls	Test	ATP
4.2.2	Power supply	Demonstration	V2TP
4.2.3	Units of Measure	Demonstration	ATP
4.2.4	Rated Output Range	Test	ATP
4.2.5	Accuracy	Test	ATP
4.2.6	Measurement Time	Test	ATP
4.2.7	Alarm Conditions	Test	ATP, V2TP
4.3	Environmental Conditions		
4.3.1	Operating Temperature	Test	V2TP
4.3.2	Storage Temperature	Test	V2TP
4.4	Usability		
4.4.1	Display Size	Inspection	V2TP
4.4.2	Measuring Site	Demonstration	V2TP

Table-1: Verification/Validation matrix.